Beginnings Of Quantum Idea:

The idea of quantization of energy was first proposed by Max Planck in context of trying to explain black body radiation. Recall from the chapter on heat transfer, how ultraviolet catastrophe gave rise to the proposal that oscillators in matter could emit energy only in multiples of $h\nu$ where ν was the oscillation frequency of the oscillator.

Photon And Quantum Theory Of Light:

In the chapter on wave optics, we emphasized on wave nature of light, however there exist phenomena such as photoelectric effect and compton scattering which could not be explained using wave theory. To explain such phenomena, Einstein developed a particle theory of light and called the quanta of light as photons. The nature of light is still being researched and phenomena like relativity or constancy of speed of light in vacuum appear very counter intuitive, however anything supported by experimental evidence has to be accepted and best efforts have to be made for developing a theoretical basis of the experiment.

According to Einsteins photon theory:

1. Light is composed of energy carrying particles (known as photons)
2. The energy of any photon is given as $\boxed{E = h\nu}$ where ν is the frequency of associated EM wave.
3. Momentum of each photon is given as $\boxed{p = \frac{h}{\lambda} = \frac{h\nu}{c} = \frac{E}{c}}$
4. Light can only be absorbed or emitted in integer multiples of $h\nu$.
5. In photon particle collisions energy and momentum are conserved but not the number of photons (photons can be absorbed and new ones emitted)
6. Photons are not deflected by EM fields since they are neutral.

→ Since photons carry energy and momentum, they are capable of exerting pressure / force on reflecting and refracting surfaces and we discussed such problem in great detail in the chapter on EM Waves. The student is advised to review the ideas of radiation pressure and force on various types of surfaces from that chapter.

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Example : NCERT :

Example 11.1 Monochromatic light of frequency 6.0×10^{14} Hz is produced by a laser. The power emitted is 2.0×10^{-3} W. (a) What is the energy of a photon in the light beam? (b) How many photons per second, on an average, are emitted by the source?

Solution: (a) $E = h\nu = 6.6 \times 10^{-34} \times 6 \times 10^{14} = 3.98 \times 10^{-19}$ Joules ≈ 2.49 eV ($1eV = 1.6 \times 10^{-19}$ Joules). Now if n is number of photons emitted per second then $P = nE \Rightarrow n = P/E = 2 \times 10^{-3} / 3.98 \times 10^{-19} \approx 5 \times 10^{15}$ photons per second.

Work Function And Electron Emission : We know that metals have free electrons which are responsible for electrical and thermal conduction, however these electrons are not so free that they would spontaneously come out of surface of the metal. There is certain amount of energy that is required to bring an electron out of the metal surface. The amount of energy required to bring an electron out of the metal surface is called the work function of the metal.

Table 11.1 Work functions of some metals

Metal	Work function ϕ_0 (eV)	Metal	Work function ϕ_0 (eV)
Cs	2.14	Al	4.28
K	2.30	Hg	4.49
Na	2.75	Cu	4.65
Ca	3.20	Ag	4.70
Mo	4.17	Ni	5.15
Pb	4.25	Pt	5.65

The work functions of some of the metals are mentioned in the table above (excerpted from NCERT). Here it is worth remembering that Alkali metals in general have low work functions with least being for Caesium (2.14eV) and largest work function is for platinum

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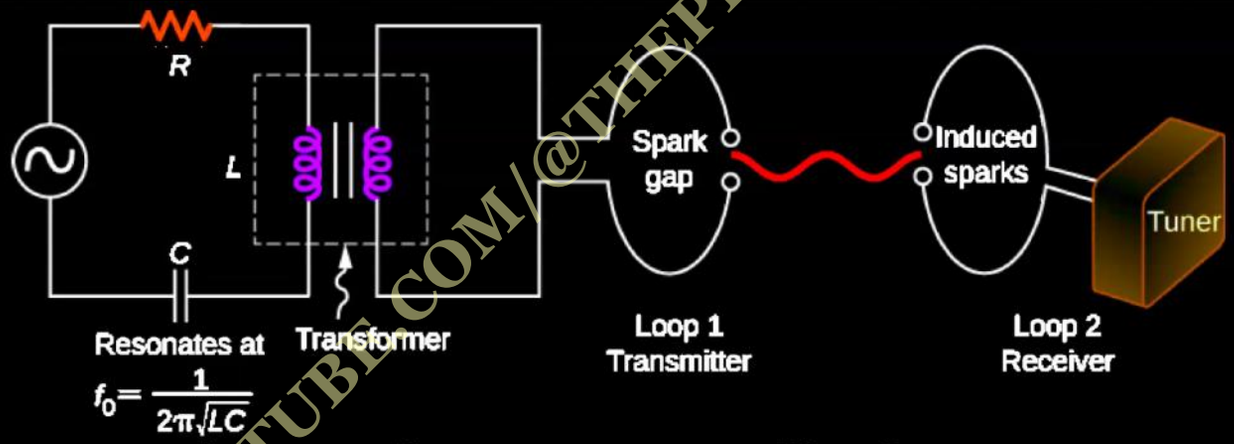
i.e 5.65 eV.

The minimum energy required to emit an electron can be provided in one of the three physical processes:

1. **THERMIONIC EMISSION**: By suitably heating a metal, sufficient thermal energy can be imparted to the free electrons so as to enable them to come out of metal surface.
2. **FIELD EMISSION**: By applying a very strong electric field (of the order of 10^8 V/m) to a metal, electrons can be pulled out of the metal (such as in spark plugs in automobiles).
3. **PHOTOELECTRIC EMISSION**: When light of suitable frequency illuminates a metal surface, electrons are emitted. These light generated electrons are called photo-electrons.

DISCOVERY OF PHOTOELECTRIC EFFECT:

HERTZ'S OBSERVATION:

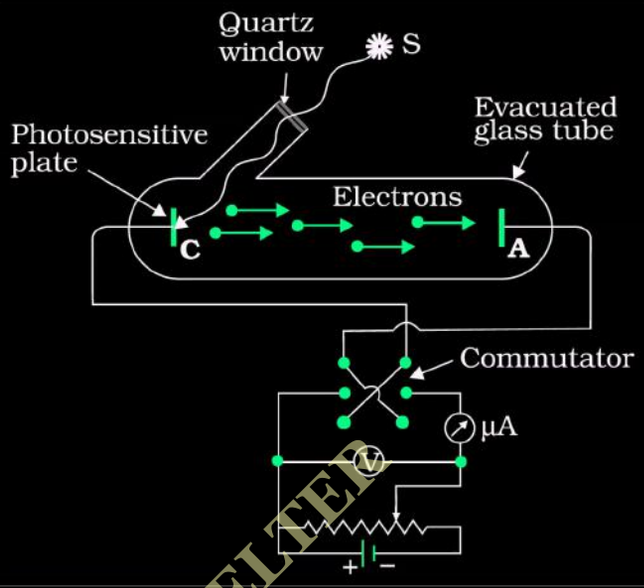


The apparatus used by Hertz in 1887 to generate and detect electromagnetic waves.

The phenomenon of photoelectric effect was first discovered in 1887 by Heinrich Hertz, during his experiments on EM Waves by means of spark discharge (schematic in the figure above). He observed that high voltage sparks across the transmitter loop were enhanced when the emitter plate was illuminated by ultraviolet light from an arc lamp. Now we understand this phenomenon quite clearly. When light falls on a metal surface, some electrons absorb sufficient energy to overcome the attraction of nearby positive ions, thus they are easily able to escape the metal surface in to surrounding space.

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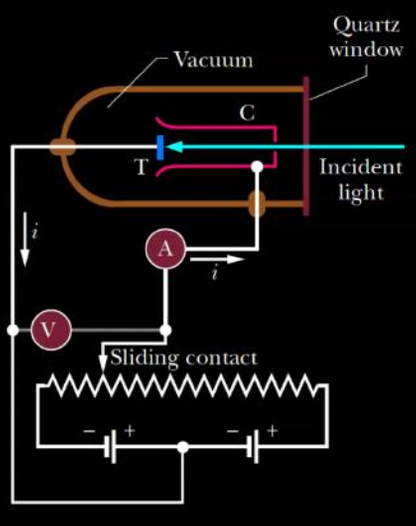
Experiments On Photoelectric Effect (Hallwach & Lenard):



Wilhelm Hallwach and Philip Lenard conducted a series of experiments to investigate the phenomenon of photoelectric effect. A schematic of their apparatus is shown in the adjacent figure. UV light is made to fall on a metal plate through a quartz window, this plate (C) emits electrons due to photoelectric effect and the emitted electrons are collected by another plate A which is maintained at some potential difference from C using an external battery whose polarity can be reversed using a commutator. The magnitude of potential difference can be adjusted using a Rheostat (variable resistance). The collected electrons can again get back to the emitter through the external circuit (battery etc). A milli ammeter and a voltmeter are provided to measure the current across the tube and the potential difference. Hallwach and Lenard performed several experiments by varying various parameters like the emitting metal, the intensity and frequency of incident light and voltage across the phototube. These experiments will be discussed in the following sections.

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Variation Of Current With Intensity And Voltage Across Tube:



Consider the adjacent figure. In this experiment, we adjust the potential difference across the photo tube by adjusting the sliding contact. Initially the collector C is kept at slightly negative potential wrt target (T). This potential difference acts to slow down the electrons because as the electrons move closer to C, they experience a repulsion. Now by

slidings the contact leftwards, the potential of C is gradually made more and more negative wrt target until the repulsion is strong enough so that readings of Ammeter becomes zero. The value of the potential of C wrt T at this point is called stopping potential.

STOPPING POTENTIAL : The reverse potential difference across the phototube that makes the photocurrent (flow of electrons due to photoelectric effect) zero is called stopping potential. At this potential even the most energetic photoelectrons lose all of their KE due to negative work done by the repelling field. Thus, if K_{max} represents KE of most energetic electron then:

$$K_{max} = eV_{stop}$$

V_{stop}: Stopping potential.

Now if ν is the frequency of incident light and ϕ is the work function of the metal then $K_{max} = h\nu - \phi$ (by energy conservation). thus:

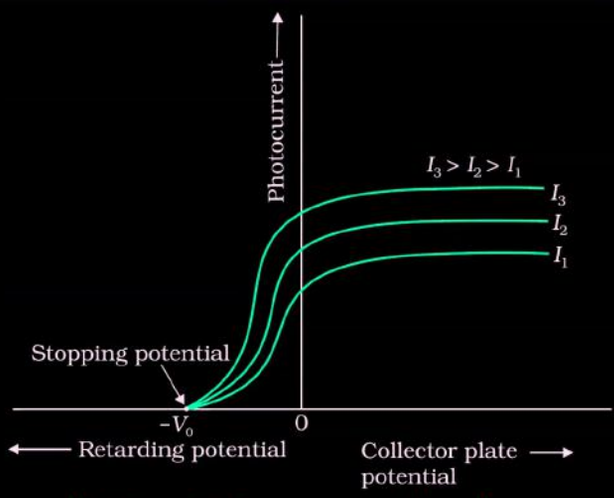
$$h\nu - \phi = eV_{stop}$$

Note that K_{max} does not depend on intensity of light, rather it only depends on frequency for same frequency, it does not matter whether light is dazzling or dim. This seems counter intuitive to the classical wave theorists because a dazzling light means a larger amplitude of electric field so one might classically expect a greater KE at a higher intensity for the same frequency, but that is not actually observed, thus photoelectric effect can't be explained in terms of classical wave theory and requires visualization of light in terms of quanta (packets) of energy.

Since K_{max} does not depend on intensity of light but only on frequency of light and work function of target metal, therefore, stopping potential also does not depend on intensity and only depends on work function and frequency of incident light.

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Saturation Current : As the potential of the collector is increased (towards zero and then positive) from stopping potential the photocurrent for a fixed intensity of light keeps increasing up to a certain value and after that there is no significant further increase in current upon increasing the collector potential. The maximum photocurrent thus obtained is called saturation current. (See figure).



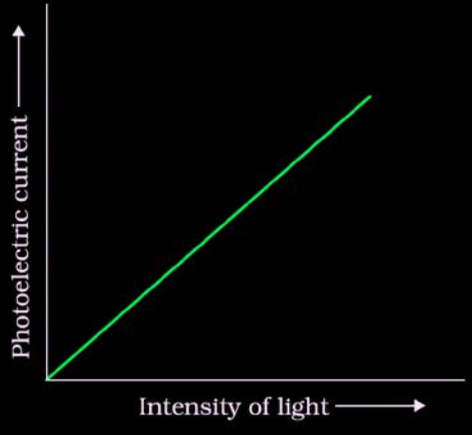
Variation of photocurrent with collector plate potential for different intensity of incident radiation.

The saturation current is directly proportional to the intensity of incident light. This is because more the number of incident photons, more the probability of hitting an electron to impart it the required energy. When the voltage of the collector is not positive enough, some of the emitted electrons might fall back again in to the emitter due to repulsion between electrons within the phototube, however when the attraction towards collector is strong enough, all the emitted electrons will start reaching the collector. Further increasing the voltage will not increase the current any further because the emission depends on intensity of light and not the phototube potential. The variation of photocurrent with tube voltage for various intensities (I) of light is shown in the graph above.

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Photoelectric Efficiency And Variation Of Saturation Current With Intensity :

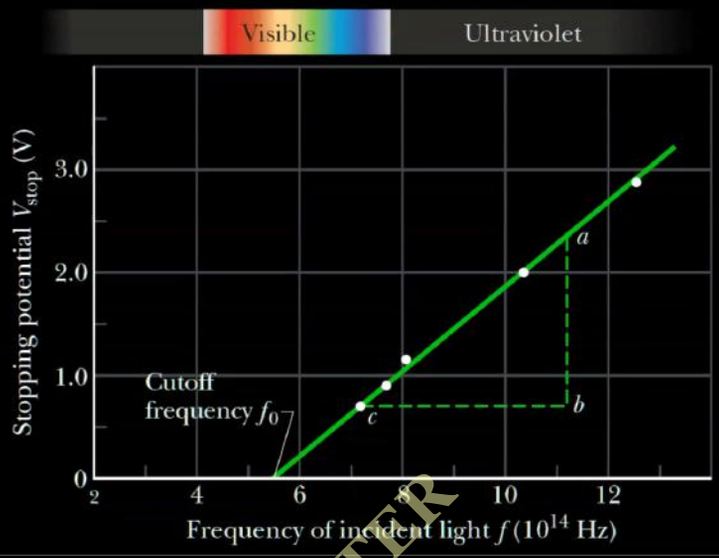
Photoelectric Efficiency : Number of emitted photoelectrons per unit incident photon is called photoelectric efficiency. Photoelectric efficiency depends on the material, surface finish and frequency of the incident light, and does not depend on intensity. Thus the saturation current is directly proportional to the intensity of the incident light (see adjacent graph).



Variation of Photoelectric current with intensity of light.

Variation Of Stopping Potential With Frequency:

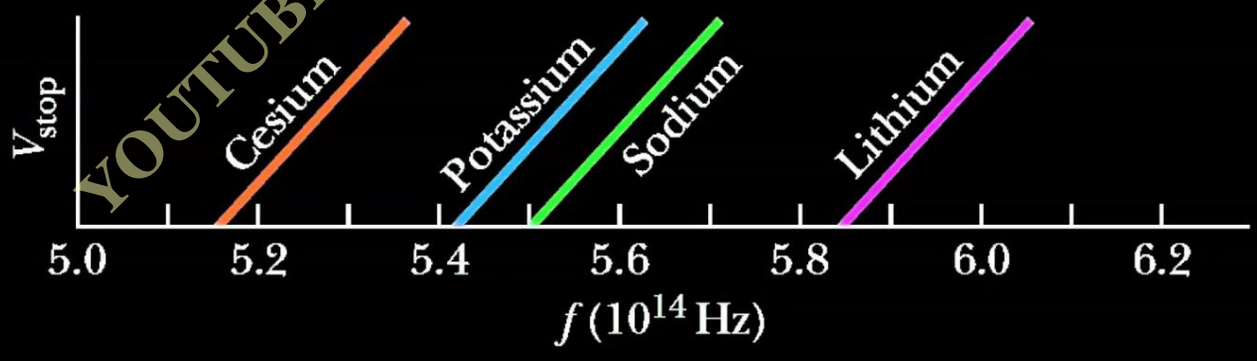
In this experiment, frequency of incident light is varied and voltage is adjusted to make the photocurrent just zero (i.e stopping potential). Here it is observed that graph of stopping potential vs incident frequency is a straight line. This is theoretically understood as follows:



$$eV_s = h\nu - \phi$$

$$\Rightarrow V_s = \frac{h}{e}\nu - \frac{\phi}{e}$$ this is of the form $y = mx + c$. Thus, the slope of V_s Vs ν graph is h/e and intercept on Vertical axis is $-\phi/e$.

Cut Off Frequency: It is observed that below a certain threshold frequency, there is no photoelectric emission. The least frequency for which photoelectric emission can occur is called cut-off frequency. This is because the photons must have an energy atleast equal to the work function so that photoelectric emission may be possible. Thus if ν_0 is cutoff frequency then $h\nu_0 = \phi$ or $\boxed{\nu_0 = \phi/h}$

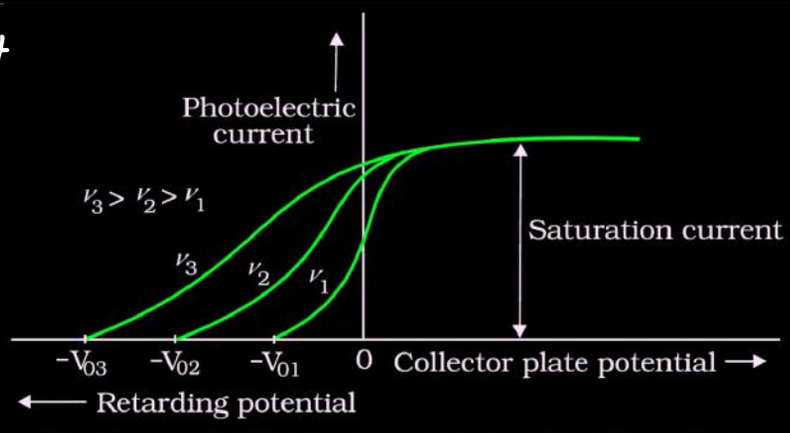


The graph of frequency vs stopping potential for various Alkali metals is shown in the figure above. For same frequency, it is seen that caesium has largest stopping potential because it has least work function. (So for same energy photons, electrons will have a larger K_{max}).

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Variation Of Photocurrent At Constant Intensity With Varying Frequency:

The graph of photocurrent at constant intensity with varying retarding potentials is as shown in the figure. It is seen that if intensity is fixed then saturation current for a given target remains constant. At points before attainment of saturation current, the current is higher for greater frequency for same retarding potential. The graph from NCERT suggests that with increase in frequency there must be an improvement in photoelectric efficiency because a smaller number of photons at higher frequency are still able to emit same number of photoelectrons at saturation.



Variation of photoelectric current with collector plate potential for different frequencies of incident radiation. (NCERT)

There seems to be some disagreement in teaching community about interpretation of word constant intensity to mean constant number of photons per second, (This confusion arises due to one of the JEE questions). However the language of NCERT is also corroborated by acclaimed modern physics texts like Arthur Beiser for example. The adjacent figure is taken from Arthur Beiser's concepts of Modern physics and it clearly says that when retarding potential is zero, photo current is same for light of a given intensity regardless of its frequency. (Though here graph is slightly different from NCERT). For JEE purposes, we will consider NCERT graph only as the correct one.

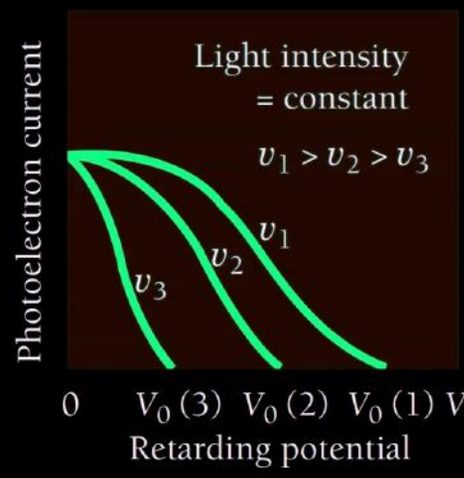


Figure 2.11 The stopping potential V_0, and hence the maximum photoelectron energy, depends on the frequency of the light. When the retarding potential is $V = 0$, the photoelectron current is the same for light of a given intensity regardless of its frequency.

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Example: Irodov 5.273: Find the threshold wavelength for zinc. Also find maximum velocity of photoelectrons emitted from its surface by light of wavelength $\lambda = 250\,nm$. Given ϕ for zinc is $3.74\,eV$. ($hc = 1242\,eV\,nm$).

Solution: $\phi = hc/\lambda_R \Rightarrow \lambda_R = hc/\phi = 1242\,eV\,nm / 3.74\,eV \approx 332.1\,nm$ Ans.

Now $K_{max} = \frac{hc}{\lambda} - \phi = \frac{1242\,eV\,nm}{250\,nm} - 3.74\,eV \approx 1.228\,eV$

Now rest mass energy of electron $= mc^2 = 511\,keV$, thus,

$$K_{max} = \frac{1}{2} m v_{max}^2 = \frac{1}{2} mc^2 \left(\frac{v_{max}}{c}\right)^2 \Rightarrow \frac{v_{max}}{c} = \sqrt{\frac{2K_{max}}{mc^2}}$$

$$\Rightarrow v_{max} = c\sqrt{\frac{2K_{max}}{mc^2}} = 3\times10^8 \sqrt{\frac{2\times1.228\,eV}{511\,keV}} \approx 6.6\times10^5\,ms^{-1}$$ Ans.

→ $hc = 1242\,eV\,nm$ $\quad$ $m_e c^2 \approx 511\,keV \approx 0.51\,MeV$

Example: Irodov 5.274:

Illuminating the surface of a certain metal alternately with light of wavelength $\lambda_1 = 0.35\,\mu m$ and $\lambda_2 = 0.54\,\mu m$, it was found that the corresponding maximum velocities of photoelectrons differ by a factor $\eta = 2.0$. Find the work function of that metal.

Solution: $K_{max} = hc/\lambda - \phi \Rightarrow \frac{v_1}{v_2} = \eta = \sqrt{\frac{hc/\lambda_1 - \phi}{hc/\lambda_2 - \phi}}$

$$\Rightarrow \phi = \frac{\eta^2 hc/\lambda_2 - hc/\lambda_1}{\eta^2 - 1} = \frac{4\times 1242/540 - 1242/350}{3}\,eV = 1.88\,eV$$

Thus $\phi = 1.88\,eV$ Ans.

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Example: Irodov 5.275:

Up to what maximum potential will a copper ball, remote from all other bodies, be charged when irradiated by electromagnetic radiation of wavelength $\lambda = 140$ nm? Given $\phi_{Cu} = 4.47\,eV$

Solution: As photoelectrons leave the copper sphere, it starts getting a positive charge and hence retarding the photoelectrons emitted later on. When the retarding potential is large enough, it will make the KE of most energetic photoelectrons zero by the time they reach infinity and then they will start falling back to the sphere. Thus maximum potential of copper sphere will be same as the stopping potential.

Thus $hc/\lambda - \phi = eV_s \Rightarrow V_s = \frac{hc}{\lambda e} - \phi/e = \frac{1242}{140} - 4.47\,V$

Thus $V_s = 4.39$ Volts Ans.

Example: Irodov 5.276:

Find the maximum kinetic energy of photoelectrons liberated from the surface of lithium by electromagnetic radiation whose electric component varies with time as $E = a(1 + \cos \omega t)\cos \omega_0 t$, where a is a constant, $\omega = 6.0 \times 10^{14}\ s^{-1}$ and $\omega_0 = 3.60 \times 10^{15}\ s^{-1}$. Given $\phi_{Li} = 2.39\,eV$

Solution: The given radiation is a mix of various frequency components which can be found by expressing the product as a sum of simple sinusoids. The maximum KE will be governed by the highest frequency term in the sum. Thus:

$$E = a\cos\omega_0 t + a\cos\omega t\cos\omega_0 t = a\cos\omega_0 t + \frac{a}{2}\left[\cos(\omega+\omega_0)t + \cos(\omega-\omega_0)t\right]$$

Thus $K_{max} = h\left(\frac{\omega+\omega_0}{2\pi}\right) - \phi_{Li}$, $h \approx 4.14 \times 10^{-15}\ eV/Hz$

thus $K_{max} \approx 4.14 \times 10^{-15} \times \frac{(3.6 + 0.6) \times 10^{15}}{2\pi} - 2.39 = 0.38\,eV$ Ans.

→ $h \approx 4.14 \times 10^{-15}\ eV/Hz$.

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Example : 5.277 :

Electromagnetic radiation of wavelength $\lambda = 0.30\,\mu m$ falls on a photocell operating in the saturation mode. The corresponding spectral sensitivity of the photocell is $J = 4.8$ mA/W. Find the yield of phtoelectrons, *i.e.*, the number of photoelectrons produced by each incident photon.

Solution: Let η be the photoelectric yield (or efficiency). Considering an incident power of 1W, number of photons incident per second, $n_p = \frac{P}{hc/\lambda}$, Thus no of electrons/sec $= \eta n_p$ and current

$i = \eta n_p \times e$ thus $J = \frac{\eta P \lambda}{hc} e \Rightarrow \eta = \frac{hc}{\lambda} \cdot \frac{J}{e} \times P \quad (P=1)$

thus $\eta = \frac{hc}{\lambda} \cdot \frac{J}{e} = \frac{1242\ eV\,nm}{300\ nm} \times \frac{4.8\ mA}{W} \cdot \frac{1}{e} = 19.8 \times 10^{-3}$

thus $\eta \approx 0.02$ or 2 in 100. Ans.

Failure Of Classical EM Wave Theory In Photoelectric Effect :

→ Within limits of experimental accuracy (10^{-9} sec) there is no time interval between arrival of light and emission of photoelectrons. A detectable photocurrent results when 10^{-6} W/m^2 of EM energy is absorbed by the sodium surface. A 1 atom thick 1m^2 sodium surface contains about 10^{19} atoms, thus each atom would recieve energy at 10^{-25} W and at this rate it will take one month to accumulate the required energy to overcome the work function.

→ A brighten light of same fuquency does not increase KE of most energetic electrons, though it increases the number of electrons while accordings to classical theory, electron energies should increase with light of greater amplitude of Electric field.

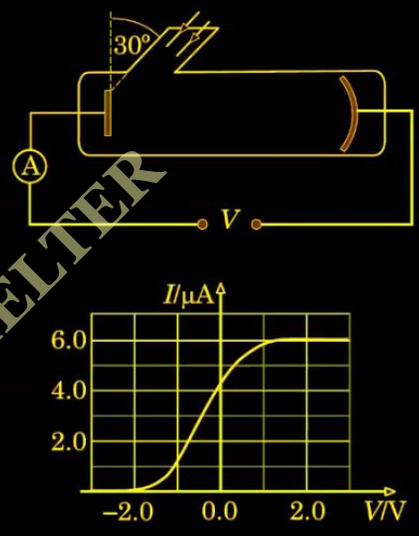
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→ According to classical theory, photoelectric emission should occur at all frequencies if the intensity is large enough however, we find that there is no photoelectric emission below a certain cut-off frequency.

EXAMPLE: PATHFINDER MCQ 13-15:

Consider a classic experiment on the photoelectric effect in which a monochromatic plane wave of light falls on a metal plate of area 66.3 cm^2 at an angle 30° as shown in the figure. Electric field in the light wave varies according to equation $E = E_0\cos(\omega t + \varphi_0)$, where amplitude is $E_0 = 15$ V/m, angular frequency is $\omega = 9.5 \times 10^{15}$ rad/s and φ_0 is some initial phase angle. For a plane wave, the density u of electromagnetic energy is given by the equation $u = \frac{1}{2}\varepsilon_0 E_0^2$. Dependence of photoelectric current I on voltage V applied between the electrodes is shown by a graph.

Speed of light is $c = 3.0 \times 10^8$ m/s, Planck's constant is $h = 6.63 \times 10^{-34}$ J·s, electrical permittivity of free space is $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m and charge on an electron is $e = -1.6 \times 10^{-19}$ C.



Q13:

What is quantum yield in this experiment?

(a) 0.019 (b) 0.038

(c) Unity when saturation current flows otherwise less than unity.

(d) It has no unique value as it depends on photoelectric current registered.

Solution: $I = uC = \frac{1}{2}\epsilon_0 E_0^2 c$, $P = \vec{I}\cdot\vec{A} = IA\cos\theta$ where θ is angle between area normal and incident light, for our case $\theta = 90° - 30°$ i.e 60°. Thus, $P = \frac{1}{2}\epsilon_0 E_0^2 c \times \cos\theta A$. Now photons / sec, $n_p = P/h\nu$

i.e $n_p = \frac{1}{2}AE_0 E_0^2 c\cos\theta \cdot \frac{2\pi}{h\omega}$, now saturation current, $i_s = \eta n_p e$

or $\eta = \frac{i_s}{n_p e} = \frac{i_s h\omega}{\pi\epsilon_0 E_0^2 AC\cos\theta e} = \frac{6\times10^{-6}\times6.63\times10^{-34}\times9.5\times10^{15}}{\pi\times8.85\times10^{-12}\times225\times3\times10^{8}\times\frac{1}{2}\times1.6\times10^{-19}\times66.3\times10^{-4}}$

≈ 0.038 option (b) Ans.

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Q14.

If amplitude of electric field in the incident light is made 25 V/m, how much will the saturation current become?

(a) 8.00 μA
(b) 10.00 μA
(c) 16.67 μA
(d) None of these

Solution: $i_s \propto$ Intensity thus saturation current will get scaled up by a factor of $(25/15)^2$ ($\because I \propto E_0^2$) thus new saturation current, $i_s' = 6\mu A \times 25/9 = 16.67\mu A$ option (c) ans.

Q15.

Instead of a simple plane wave, a complex plane wave of light governed by the following equation falls on the metal plate.

$$E = E_0\left\{1+\cos\left(\omega t+\varphi_0\right)\right\}\cos\left(\omega t+\varphi_0\right)$$

Stopping potential and saturation current respectively are

(a) 6.27 V and 6.75 μA
(b) 6.27 V and 7.50 μA
(c) 8.27 V and 6.75 μA
(d) 8.27 V and 7.50 μA

Assume Quantum efficiency to be independent of frequency.

Solution: $E_0(1+\cos(\omega t+\phi_0))\cos(\omega t+\phi_0)$

$= E_0\cos(\omega t+\phi_0) + \frac{E_0}{2}\left[\cos(2\omega t+2\phi_0)+\cos 0\right]$

$= \frac{E_0}{2} + E_0\cos(\omega t+\phi_0) + \frac{E_0}{2}\cos(2\omega t+2\phi_0)$

(I) (II) (III)

The radiation can be broken in to three components as shown above, (I) is constant field so produces no current (II) is the original field which produces 6μA, intensity of (3) is $\frac{1}{4}$ that of (II)

So it would have produced $\frac{1}{4}\times 6$ i.e 1.5μA if the efficiency increased due to doubled frequency. But assuming same efficiency, the current will be 1.5/2 i.e 0.75 μA. Thus, new saturation

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current, $i_s = 6 + 0.75 = 6.75\,\mu A$

Now let V_s be the new stopping potential, then according to graph,

$$\left.\begin{aligned} h\omega - \phi &= e \times 2 \\ \text{and} \quad 2h\omega - \phi &= eV_s \end{aligned}\right\} \Rightarrow h\omega = e(V_s - 2)$$

$$\Rightarrow V_s = \frac{h\omega}{e} + 2 = 4.14 \times 10^{-15}\ eV/Hz \times \frac{9.5 \times 10^{15}\ Hz}{2\pi} \cdot \frac{1}{e} + 2$$

$$= 8.26\ V$$

Thus best option: (C).

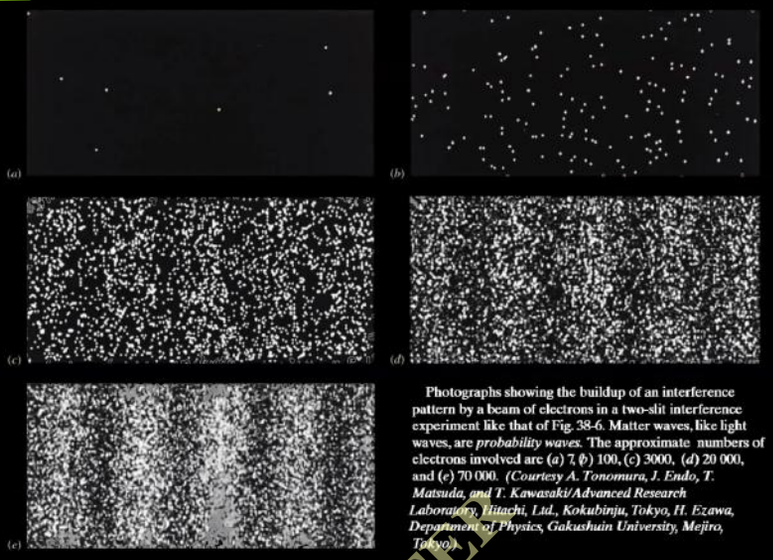
→ Note that this question should have mentioned that quantum efficiency is to be taken as constant, otherwise as per NCERT theory, the saturation current will be $6 + 1.5$ i.e $7.5\,\mu A$ and correct option will be (D) then.

WAVE NATURE OF MATTER: Just as light exhibits both particle and wave nature, with wavelength being related to momentum as $\lambda = h/p$ similarly De-Broglie proposed that particles also exhibited wave nature with wavelength of the matter waves also given as $\boxed{\lambda = h/p}$. De Broglie had just proposed this idea based on symmetry argument for nature and initially his proposition was not well recieved however later on experiments were conducted which confirmed De Broglie's hypothesis namely ① Electron diffraction through circular apertures ② Davisson and Germer's experiment ③ Double slit experiments using electrons.

If a wavelength is proposed for material particles then there must also be a wave associated with a particle. Schrodinger worked on the idea and wrongly thought that particles were smeared out in space however later on Max Born gave the correct interpretation to the De Broglie wave as complex probability waves whose square of magnitude represented instantaneous probability of finding a particle.

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Two Slit Electron Interference Pattern:

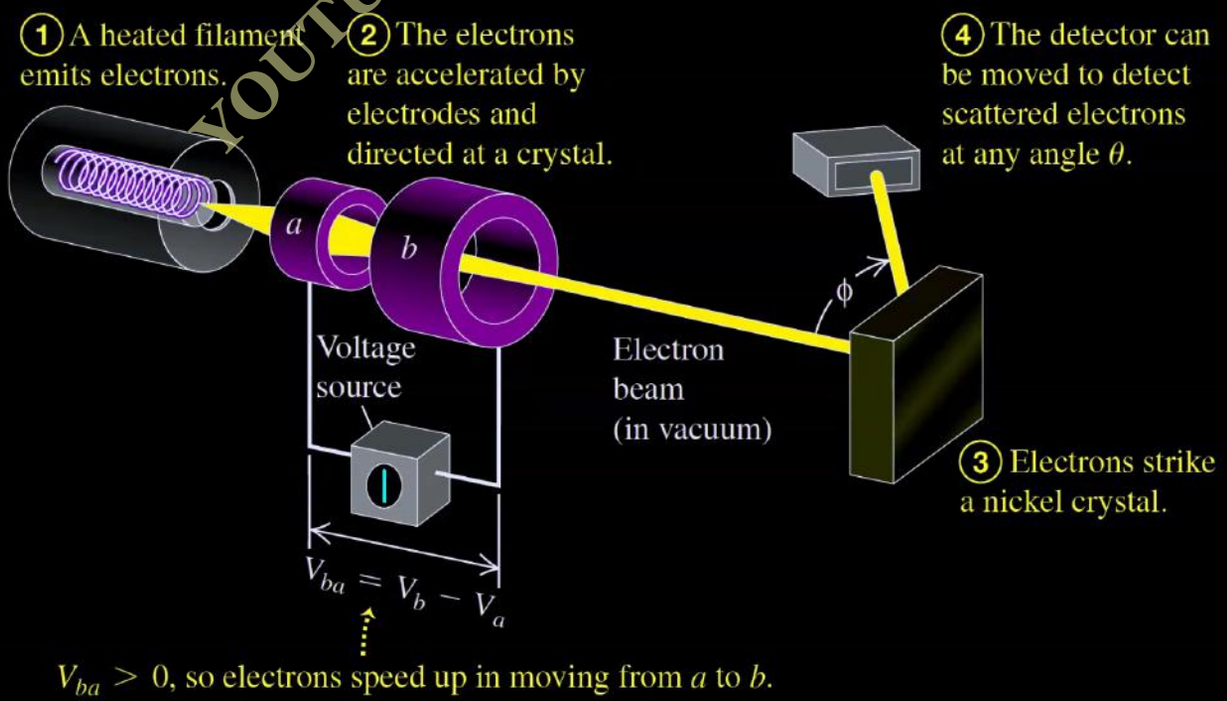


Photographs showing the buildup of an interference pattern by a beam of electrons in a two-slit interference experiment like that of Fig. 38-6. Matter waves, like light waves, are *probability waves*. The approximate numbers of electrons involved are (*a*) 7, (*b*) 100, (*c*) 3000, (*d*) 20 000, and (*e*) 70 000. *(Courtesy A. Tonomura, J. Endo, T. Matsuda, and T. Kawasaki/Advanced Research Laboratory, Hitachi, Ltd., Kokubinju, Tokyo, H. Ezawa, Department of Physics, Gakushuin University, Mejiro, Tokyo.)*

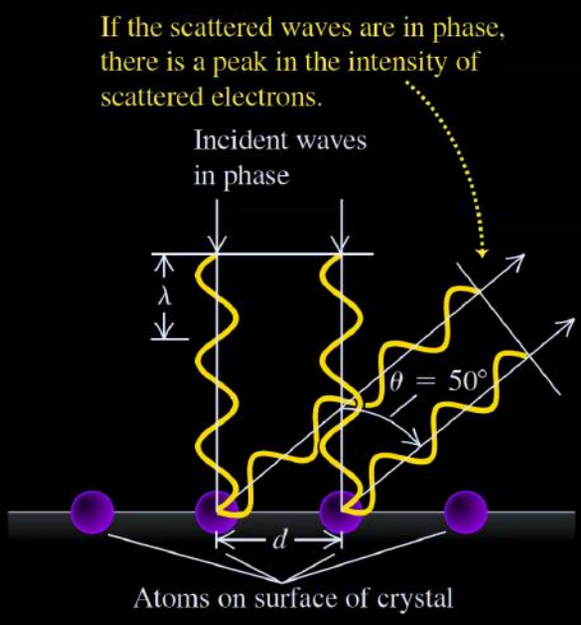
The adjacent figure shows the outcome of a double slit electron interference pattern using a set up similar to YDSE. With very few electrons we don't see any clear pattern, rather we get some random spots, however as more and more electrons hit the screen a pattern of maxima and minima begins to emerge. The density of spots being governed by the interference of probability waves from two slits.

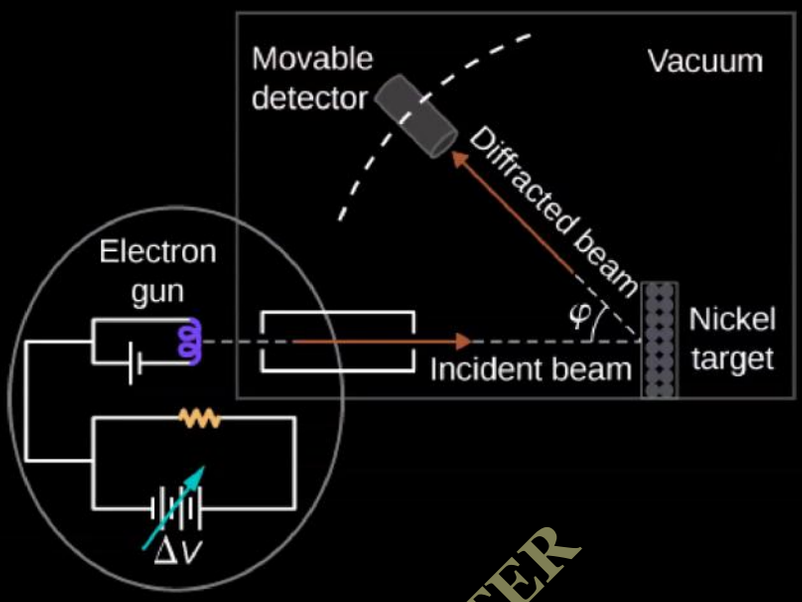
Similar experiments have also been done with photons. With classical wave theory we do not visualise light as photons but one can still think "what will happen on the screen if a single photon were incident on the double slit apparatus?". It turns out that with photons also the experiments have been carried out with extremely dim intensity and same picture emerges as with interference pattern of electrons.

Davisson And Germer Experiment:

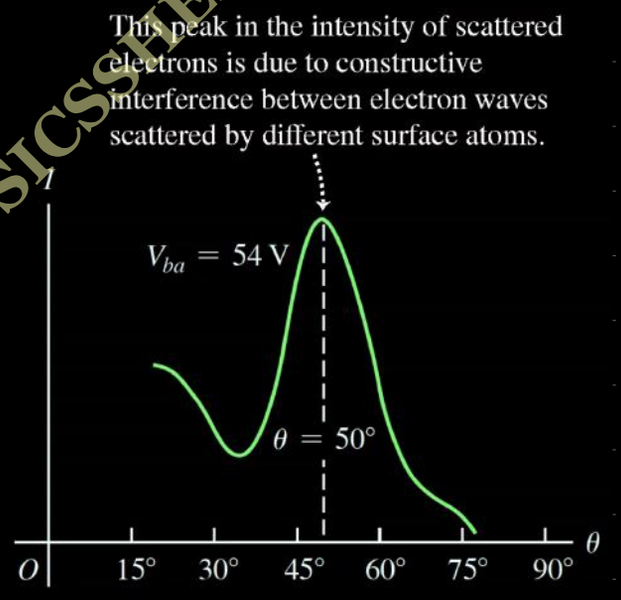


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Davisson and Germer were studying diffraction of Electrons from polycrystalline nickel. A lucky accident changed the entire direction of their research. Due to the accident, air entered their apparatus and nickel surface got oxidized. They heated the nickel piece again in a suitable environment to deoxidize and upon cooling, they got a single crystal of nickel.



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SCATTERING ANGLE: Angle between incidence direction of electrons and scattering direction is called scattering angle (angle ϕ in the experiment schematic). Davisson and Germer studied the relative intensities of electrons at different scattering angles and found that for an accelerating Voltage of 54 V, the maximum intensity was obtained at a scattering angle of 50° (see the graph above). With the classical particle picture, this behaviour would seem weird, however, the surface of the nickel crystal acts like a diffraction grating with a spacing equal to separation between the surface atoms. Thus for a scattering angle ϕ, the path difference between neighboring rays is $d\sin\phi$. Thus for constructive interference, $\boxed{d\sin\phi = n\lambda}$. ← ϕ is scattering angle

d : interatomic spacing.

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The lattice spacing in nickel (d) was already known from X-ray crystallography to be $d = 2.15\,\text{Å}$ thus λ could be calculated as:

$$\lambda = d\sin 50^\circ = 2.15\,\text{Å} \times \sin 50^\circ \simeq 1.65\,\text{Å} \quad (\text{using } n=1,\ \phi = 50^\circ).$$

Now we can also calculate λ using De-Broglie's hypothesis:

$$K = eV = p^2/2m \Rightarrow p = \sqrt{2Km} \text{ thus,}$$

$$\lambda = h/p = \frac{h}{\sqrt{2Km}} = \frac{h}{\sqrt{2eVm}} = \frac{1.227}{\sqrt{V}}\,\text{nm} = \sqrt{\frac{150}{V}}\,\text{Å}$$

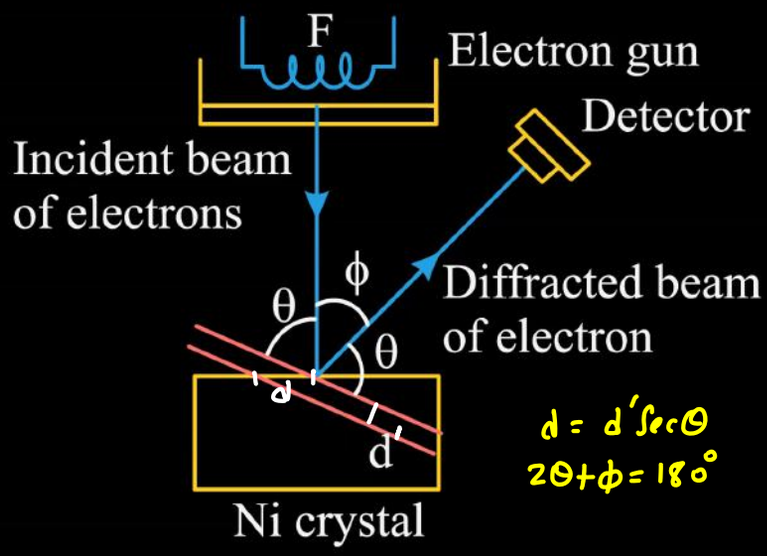
(Here V is accelerating potential). Thus for electrons:

$$\boxed{\lambda = \sqrt{\frac{150}{V}}\,\text{Å}}$$

Now using $V = 54$ Volts, $\lambda = \sqrt{\frac{150}{54}}\,\text{Å} = 1.67\,\text{Å}$ this is in very close agreement with the experimental value of $1.65\,\text{Å}$.

Glancing Angle And Bragg Planes:

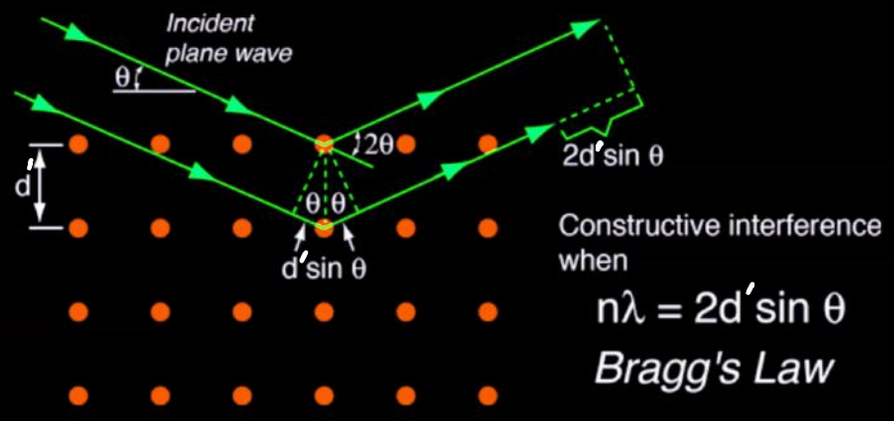
When a maxima is detected in the davisson and germer experiment, there is a family of planes containing atoms (Bragg plane) which is equally inclined to incident and diffracted beam directions. The angle which the incident on the diffracted beam makes from such Bragg Plane is called the glancing angle. If d is the interatomic spacing of the surface atoms and d' the spacing of Bragg planes, it can be shown that $d\sin\phi = 2d'\sin\theta$. Thus for constructive interference we can also write $\boxed{2d'\sin\theta = n\lambda}$, Here d' is the spacing between



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the bragg planes.

Visualizing Davisson And Germer In Terms Of Bragg Diffraction:



If d' is the spacing between Bragg planes, the Davisson and Germer experiment can also be visualized in terms of constructive interference between rays reflected from various bragg planes as shown. Thus if d' is the spacing between Bragg planes, the path difference between reflected rays from two adjacent Bragg planes is $\Delta x = 2d'\sin\theta$.

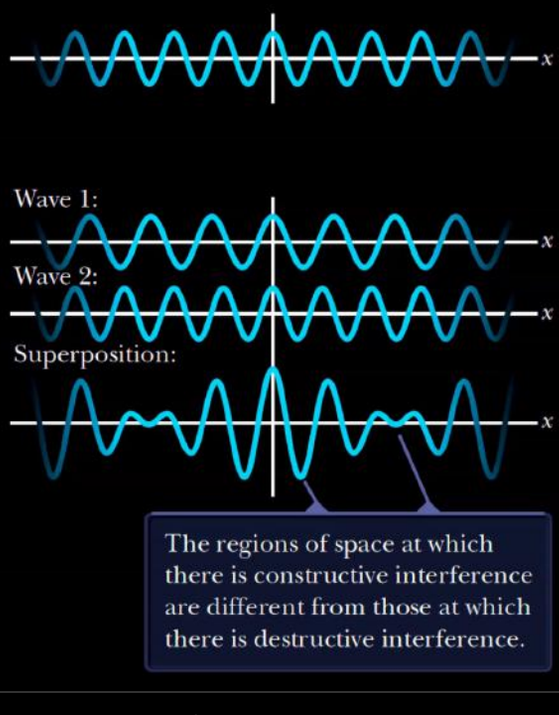
→ Thus for Davisson and Germer Experiment:

$$\boxed{d\sin\phi = 2d'\sin\theta = n\lambda}$$

d: Interatomic spacings on surface ϕ: Angle of scattering
d': Bragg planes spacing θ: Glancing Angle.

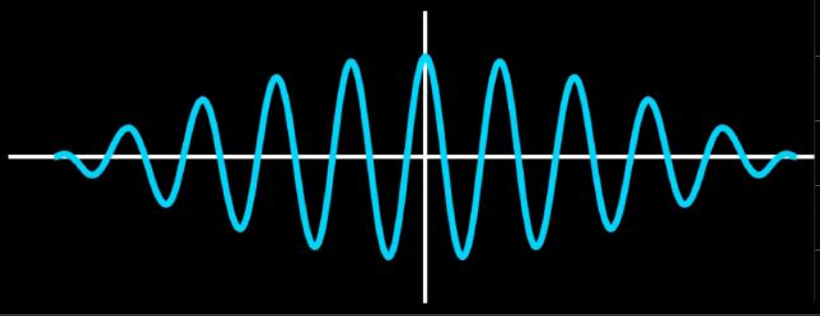
Generating A Wave Function To Represent Localized Particle:

An ideal particle has zero size and is perfectly localized in space whereas an ideal wave such as $A\sin(kx-\omega t)$ has a single frequency and exists from $-\infty$ to $+\infty$ so how can waves generate a localization. Recall that when waves of slightly different frequencies are superposed, beats are formed thus producing some kind of localization in time. Similarly when waves of slightly different wavelengths are superposed, a beat kind of phenomena is generated in space.



Now imagine superposing infinite number of waves having wave propagation constants distributed between $k \pm \Delta k$. If all such

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phasors are aligned for certain value of x then as we move a distance Δx, the phasors will be spread out in an angle $\Delta k \Delta x$ as Δx increases, the angular spread of the phasor fan will keep on increasing leading to a smaller and smaller resultants as we move outwards (just like in slit diffraction we consider infinite phasors of infinitesimal amplitude). Thus superposing infinite number of infinitesimal waves with propagation constants lying between $k \pm \Delta k$, we get practically a localized zone in which there is significant amplitude and far away from this zone, the amplitude is practically zero and hence zero probability of finding the particle. Also note that more the value of Δk, quicker the wave function will die out because larger Δk leads to quicker fanning out of phasors. Thus with increase in Δk, the uncertainty in position reduces. Thus if we denote Δx as uncertainty in position we can say that, $\Delta x \propto \frac{1}{\Delta k}$

Now $k = \frac{2\pi}{\lambda} = \frac{2\pi}{h} p \Rightarrow \Delta k = \frac{2\pi}{h} \Delta p$, thus we can say that $\Delta x \propto \frac{h}{2\pi \Delta p}$, thus more the uncertainty in momentum, lesser is the uncertainty in position. This is called Heisenberg's Uncertainty Principle.

HEISENBERG'S UNCERTAINTY PRINCIPLE: As shown in the previous section, the uncertainty in position and momentum are inversely related, this is mathematically expressed as:

$$\Delta x \, \Delta p \approx h/2\pi = \hbar$$

Here "$\approx$" should be read as "of the order of". A mathematical analysis shows that $\Delta x \, \Delta p \geq \frac{h}{4\pi} = \frac{\hbar}{2}$.

Using the symbols Δx makes one think that there is an

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upper bound to uncertainty in position, but this is not a correct statement, the wave function still exists all the way up to infinity even though its amplitude is practically zero. More advanced text books use statistical approach and talk in terms of most expected values and replace Δ by σ i.e. $\sigma_x \sigma_p \geq \frac{\hbar}{2}$, however it is beyond the scope of discussion at JEE level. (σ_x is a number generated by considering probabilities at various x, it is similar to standard deviation in discrete variables, though in quantum physics such ideas are applied to continuous functions).

De-Broglie Wavelength Of Proton Neutron And Deutron:

Recall that De-Broglie wavelength for any particle is given by

$$\lambda = \frac{h}{\sqrt{2mK}}$$ where K is kinetic energy. Many times in JEE physics problems, we come across the need to find De-Broglie wavelengths for various standard particles whose energy is known in eV (electron Volts). Putting in the mass values and charge values, if the particles are accelerated through V volts, their De-Broglie wavelengths are as follows:

$$\lambda_\alpha = \frac{0.101}{\sqrt{V}}\ \text{Å} \qquad \lambda_d = \frac{0.202}{\sqrt{V}} \qquad \lambda_p = \sqrt{2}\,\lambda_d$$

p: proton, d: deutron, α: Alpha Particle.

→ If you remember the formula for α particle you can readily infer the other formulae. For same potential difference, α particle will have double KE as well as double mass as compared to deutron while proton will have half the mass of deutron but same KE.

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Schrodinger's Equation:

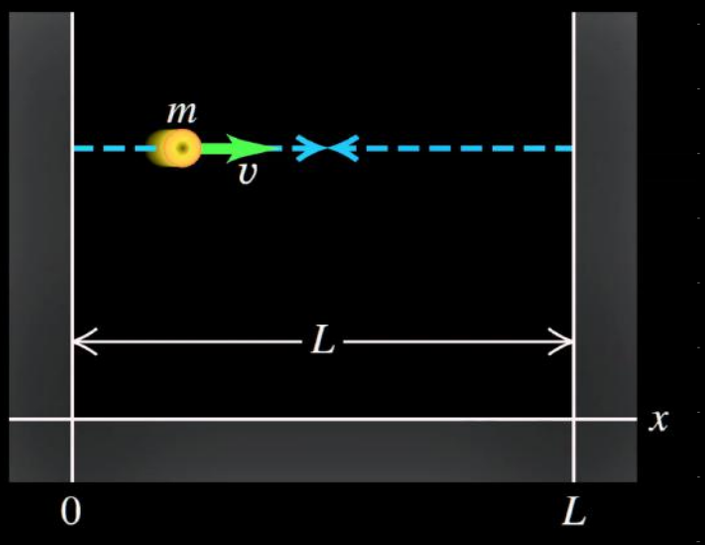
Erwing Schrodinger had proposed a wave theory describing locations of subatomic particles which was later reinterpreted by Max Born in terms of probabilities of finding a particle anywhere is space and time. A detailed discussion of Schrodinger's theory is beyond the scope of discussion at JEE level however we shall directly state the schrodinger's equation for a particle free to move along a single direction:

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2 m}{h^2}[E - U(x)]\psi = 0 \quad (1)$$

Here E is the total energy of the particle and $U(x)$ is the potential energy of a particle at a general x. Schrodinger equation cannot be derived from any fundamental principles rather it is itself a basic principle of nature. We can solve equation (1) for various kinds of potential energy functions subjected to given constraints or boundary conditions.

Particle In A Box (Infinite Potential Well):

Consider a particle confined to move along a one dimensional box from $0 < x < L$. The potential energy function is $U(x) = 0 \ \forall \ x \in (0, L)$ and ∞ everywhere else. Thus any particle with finite KE must remain trapped and the probability of finding it must be zero for $x \in (-\infty, 0] \cup [L, \infty)$. Now schrodinger equation for this particle becomes:



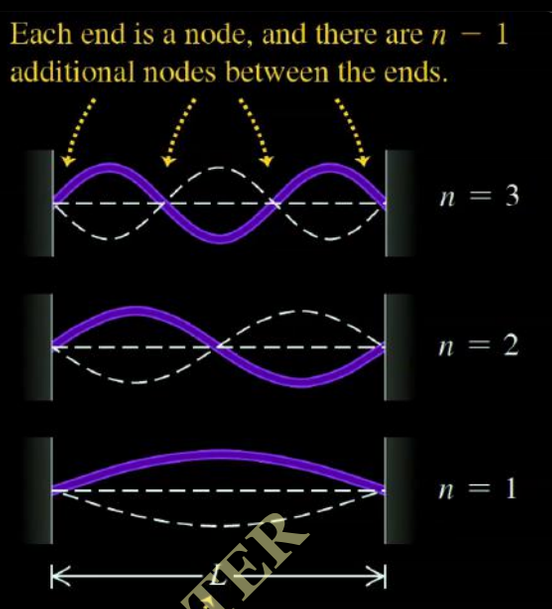
$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2 mE}{h^2}\psi = 0 \ , \text{ this is a standard SHM}$$

type of differential equation (though in x), so its solution must be

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sinusoidal. Also the value of the wave function must be zero at $x=0$ and $x=L$. This is similar to a situation in stationary waves on strings where end points must be a node. Thus possible set of wavelengths of the particle is given as $L = n\lambda/2$ which can be used to calculate permissible values of momenta as:

$$L = \frac{nh}{2p} \Rightarrow \boxed{p_n = \frac{nh}{2L}} \quad n = 1, 2, 3 \ldots$$



Now KE in the region can be found using $K = p^2/2m$ thus permissible energies of the particle:

$$\boxed{E_n = \frac{n^2 h^2}{8mL^2}} \quad n = 1, 2, 3 \ldots$$

→ According to classical intuition we might be tempted to think that any value of KE would be possible, but quantum mechanics limits our choices. However since h is very small, for massive bodies we may not even notice this quantization because the values of permissible energies will be very close together.

→ Classically we may think that there is equal probability of finding the particle everywhere from 0 to L however, according to quantum mechanics we see that wherever we have nodes, there is zero probability of finding the particle. The probability is position dependent even though KE is constant. This is totally counter intuitive but that's how it works out for sub-atomic particles. (And this is just the beginning of counter intuitive results in Quantum Mechanics).

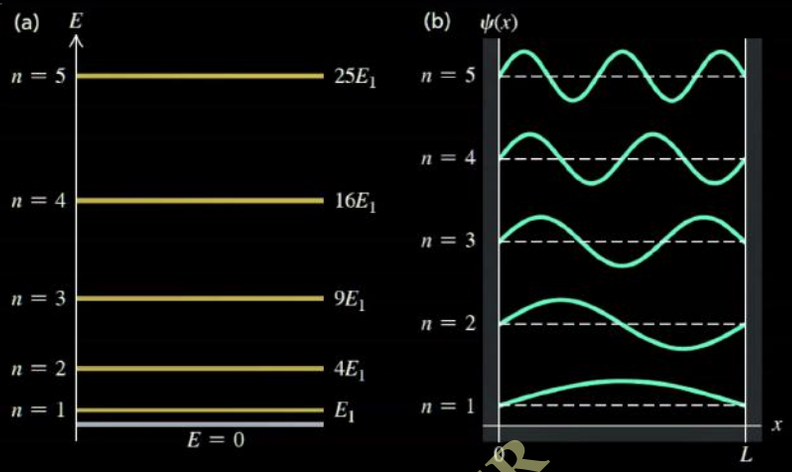
Considering the intuitive absurdity of the ideas Bohr remarked:
"If you are not confused about quantum mechanics, you have not really understood it".

→ The solution of schrodinger equation in our case is $\psi(x) = C \sin(n\pi x/L)$
Note that $n \neq 0$ since that would make ψ zero everywhere

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which would mean that particle does not exist anywhere. Thus, particle can neither have zero momentum nor zero energy.

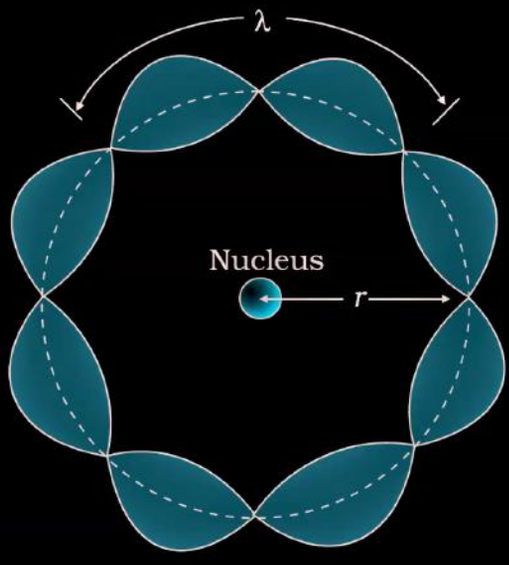
→ Some of the permissible energy levels of the particle are shown in adjacent figure.



→ If several particles are to be accomodated in the box, Pauli's exclusion principle must be followed i.e. if we are confining electrons, each level can have a maximum of two electrons with opposite spins. Using this we can generate possible energy states of the system under constraint of a given number of total particles.

Understanding Bohr's Quantization Rule In Terms Of De-Broglie Wavelength:

Exact shape of orbitals can be worked out by solving the schrodinger equation in 3D polar coordinates. Here we will only justify the Bohr Quantization rule in terms of a stationary matter wave. The probability wave around a nucleus can be modeled as superposition of two schrodinger waves (one clockwise, another anticlockwise). Thus the superposition leads to a stationary wave. Now if there is a node somewhere then diametrically opposite point must also be a node because path difference to the diametrically opposite point is zero for the two waves. Thus we need to accomodate integer number of loops in half the circumference accordingly,



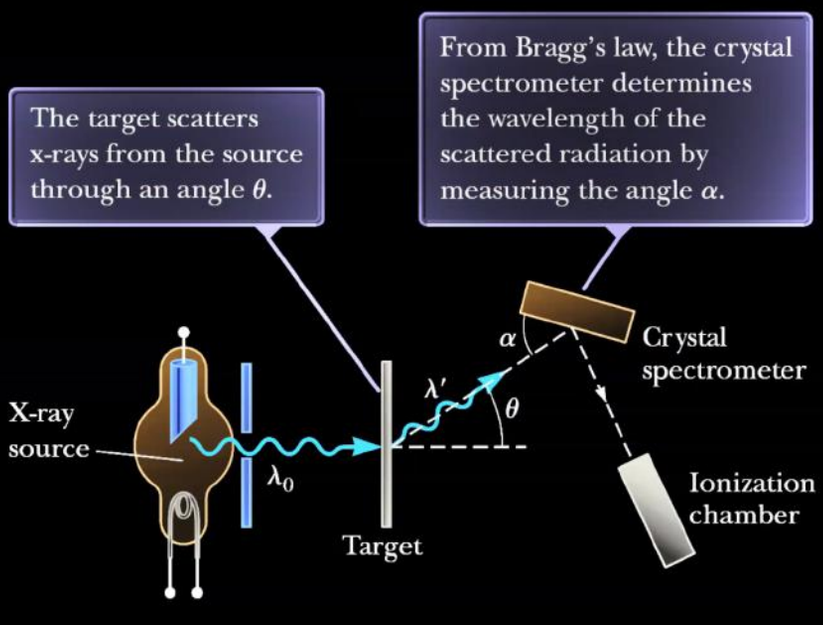
A standing wave is shown on a circular orbit where four de Broglie wavelengths fit into the circumference of the orbit.

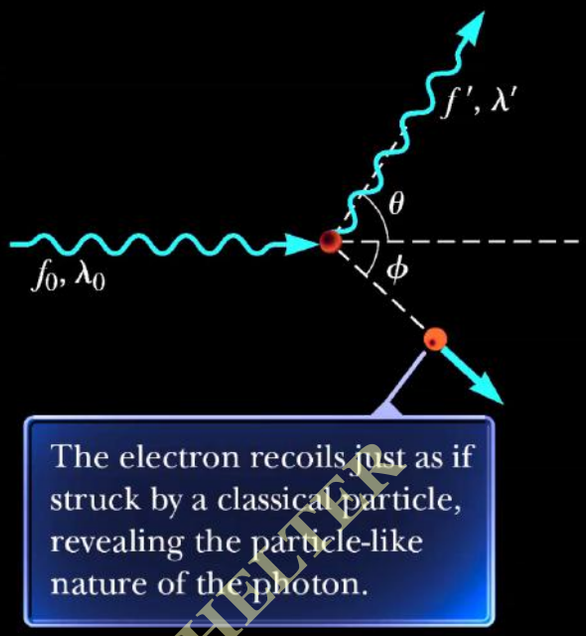
$$n(\lambda/2) = \pi r \Rightarrow \frac{nh}{2p} = \pi r \text{ or } \frac{nh}{2mv} = \pi r$$

$$\Rightarrow \boxed{mvr = \frac{nh}{2\pi}} \leftarrow \text{Angular momentum Quantization Rule.}$$

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Compton Scattering:





Apart from photoelectric effect another effect that can be explained using particle nature is the Compton effect. Here an Energetic X-Ray photon is scattered by an electron. The photon imparts some energy to the electron in the scattering process thus loses energy and therefore its frequency decreases ($\because E = h\nu$) which also means that its wavelength will increase. This increase in wavelength of a photon due to scatterings from an electron is called Compton Effect. The quantitative expression for change in wavelength during scattering can be obtained using energy and momentum conservation (considering relativistic effects). Here we omit the derivation and state the result for compton shift without proof:

$$\lambda' - \lambda_0 = \frac{h}{m_e c}[1 - \cos\theta]$$ ← (compton shift (optional)

Here λ_0 is the wavelength of incident photon, λ' is wavelength of scattered photon and m_e is mass of the electron and θ is the scattering angle of the photon. The factor $h/m_e c$ is called compton wavelength and has a value 2·43 pm.

$$\frac{h}{m_e c} = 2{\cdot}43 \text{ pm}$$

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→ A free electron can not absorb a photon, because it would require the electron to have a velocity more than c. This can be shown as follows: Let photon energy be E, then by momentum conservation, electron momentum must be $p = E/c$

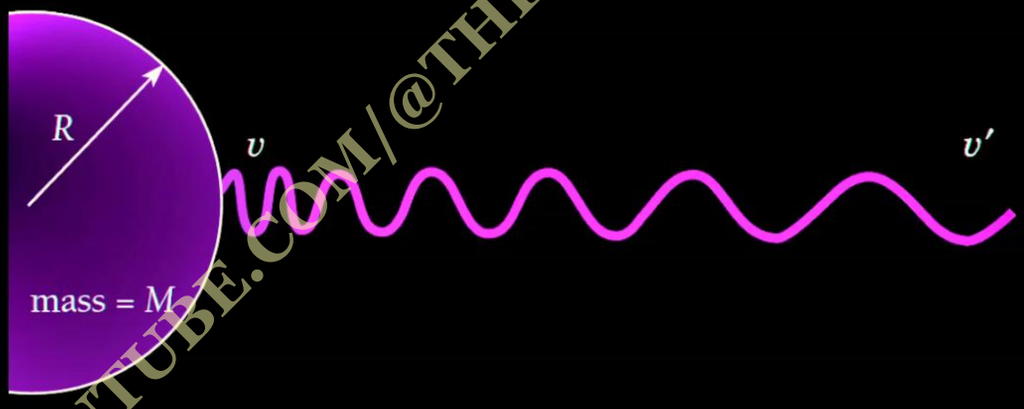
Also $E = p^2/2m \Rightarrow E = \frac{E^2}{c^2} \times \frac{1}{2m} \Rightarrow E = 2mc^2 = \frac{1}{2}mv^2$

$\Rightarrow v = 2c$ which is a contradiction since nothing can travel faster than light.

PHOTONS AND GRAVITY: Although a photon has no rest mass, for interaction with gravitational fields, a photon behaves as if it has a gravitational mass given by:

$$m = p/c = h\nu/c^2$$

GRAVITATIONAL RED SHIFT:



The frequency of a photon emitted from the surface of a star decreases as it moves away from the star.

Due to gravitational mass, photons behave somewhat like material particles in their interaction with gravitational fields. Thus as a photon recedes away from a star, it gains PE and loses the "$h\nu$", sort of KE. Thus its frequency becomes smaller and wavelength larger and larger as it recedes. Let ν be the frequency of the photon near star of mass M and let ν' be its frequency upon escaping to infinity. Then by Energy Conservation:

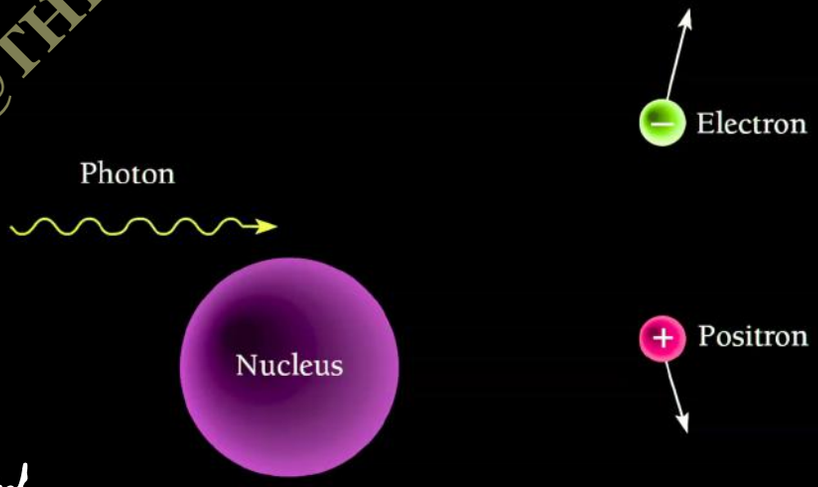
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$$-\frac{GMm}{R} + h\nu = h\nu' \Rightarrow -\frac{GM}{R}\cdot\frac{h\nu}{c^2} + h\nu = h\nu'$$

$$\Rightarrow \nu' = \nu\left[1 - \frac{GM}{Rc^2}\right]$$, this is the frequency of the photon upon escaping from gravitational field. If the term in the bracket becomes zero (or negative), photon cannot escape the star and the star is considered a black-hole. We have done a very simplistic analysis, exact analysis involves general relativity and it turns out that if $GM/Rc^2 > 1/2$, the star behaves as a blackhole and no light escapes it. The required radius for $GM/Rc^2 = 1/2$ is called Schwarzschild radius, $R_s = 2GM/c^2$. A body is considered a blackhole if all its mass lies within this radius. The escape speed from a blackhole is equal to the speed of light at schwarzschild radius thus nothing can escape a blackhole.

PAIR PRODUCTION:

If a photon has sufficiently high energy, it is possible for it to materialize in to an electron and positron pair. Here electromagnetic energy is converted in to matter according to the equation $E = mc^2$.



No conservation principles are violated in pair production (Energy, Momentum and Charge are conserved).

The energy here includes rest mass energy of the particles. The linear momentum is conserved with help of a nucleus which carries away enough photon momentum for the process to occur. Because of enormous mass, nucleus absorbs only a very small fraction of photon energy. Pair production cannot occur in free space because energy and momentum conservation could not be simultaneously satisfied.

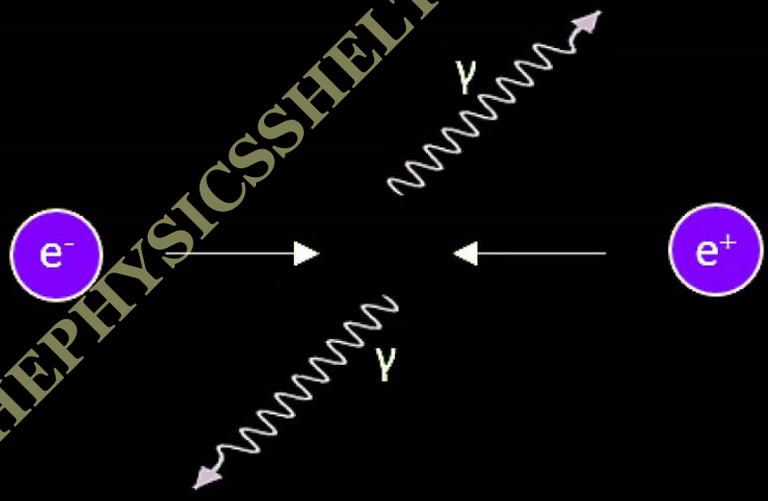
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This is because from CM frame of the pair, the momentum of the pair would be zero, which would require photon momentum to be also zero and this is not possible because zero momentum would also imply zero energy ($\because E = pc$).

The rest mass energy of an electron or positron is 0.51 MeV hence pair production requires a photon energy of atleast 1.02 MeV (i.e γ rays). Any excess photon energy would appear as Kinetic Energy of the pair.

PAIR ANNIHILATION:

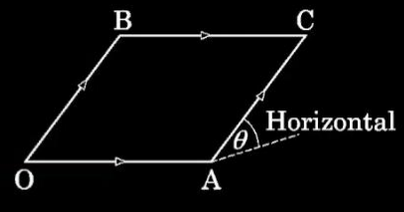
Just as a photon can materialize in to an electron positron pair similarly electron and positron can annihilate each other to produce two Gamma photons with equal and opposite momenta. The minimum energy of a Gamma photon generated in this manner is 0.51 MeV (rest mass energy of positron or electron). If there is additional Kinetic Energy, it will lead to γ photons of still higher energy.



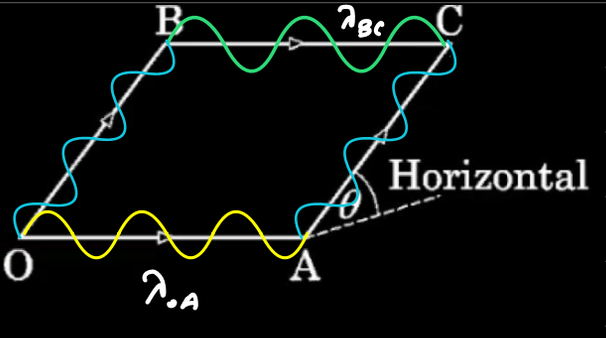
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EXAMPLE: PATHFINDER: BUILD: S:

A neutron beam of initial kinetic energy E_0 is divided into two beams at point O by some mechanism. Thereafter the beams proceed on the path OAC and OBC without a phase change at the corners A and B to interfere with each other at point C. Here OACB is a square of side l tilted at an angle θ above the horizontal. Mass of a neutron is m, Planck's constant is h and acceleration of free fall is g. Find an expression for the angle θ at which a maxima is detected at point C.



Solution: For constructive interference, there should be a difference of integer number of De Broglie wavelengths along the paths OAC and OBC. ($\because$ we want crest to fall on crest). Now at some level, KE of



neutron is same thus OB and AC contain exactly same number of De-Broglie wave cycles. Thus we want:

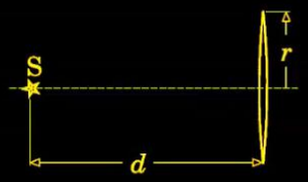
$$l/\lambda_{OA} - l/\lambda_{BC} = n \quad (1) \text{ where } n \text{ is some integer.}$$

Now $\lambda = h/\sqrt{2mE}$ thus equation (1) becomes:

$$\frac{l}{h}\left[\sqrt{2mE_0} - \sqrt{2m(E_0 - mgl\sin\theta)}\right] = n, \text{ Rearranging and solving for } \theta:$$

$$\theta = \sin^{-1}\left[\frac{2nhl\sqrt{2mE_0} - n^2h^2}{2m^2gl^3}\right], \; n = 1, 2, 3\ldots \text{ Ans.}$$

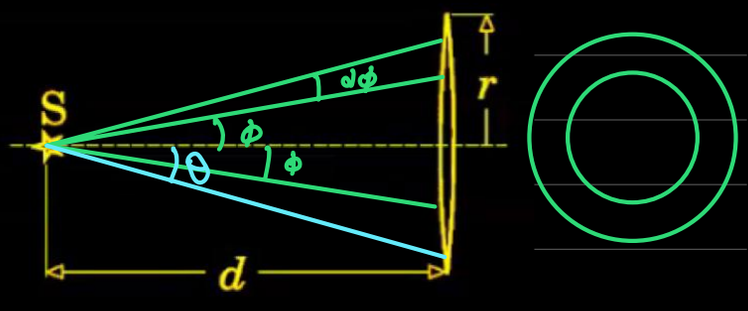
EXAMPLE: PATHFINDER: CHECK 1:



An isotropic point light source S of power P, located a distance d from a specially designed non-absorbing, non-dispersing and non-reflecting lens. This lens converges all the radiant energy coming from the source into a beam parallel to its optical axis. Find magnitude of the force of radiation pressure on the lens.

Solution: The situation is like that of variable particle system. The force on the lens must be equal to difference in outflow rate of momentum and inflow rate of momentum of the photons.

We will try to find the momentum crossing an annulus between angular radii ϕ and $\phi + d\phi$ and then integrate it from 0 to θ to get the net inflow rate of momentum.



Now Momentum per unit solid angle, per unit time:

$$f_\Omega = \frac{1}{4\pi}\left(\frac{P}{c}\right) \quad (1) \quad (\because E = pc)$$

Now recall that solid angle of a disc of semi vertex angle ϕ

$\Omega = 2\pi(1 - \cos\phi) \Rightarrow$ solid angle of annular disc, $d\Omega = 2\pi\sin\phi\, d\phi$ (2)

We just need to know component of horizontal momentum

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crossing per unit time thus: $df = f_\Omega \times d\Omega \times \cos\phi$

$\Rightarrow df = \frac{P}{4\pi c} \times 2\pi \sin\phi\, d\phi \times \cos\phi \Rightarrow f = \frac{P}{2c} \int_0^\theta \sin\phi \cos\phi\, d\phi$

$\Rightarrow f = \frac{P}{2c} \frac{\sin^2\theta}{2} = \frac{P}{4c} \sin^2\theta$ ③

Now total incident Power on lens / unit time: $P_e = \frac{P}{4\pi} \times 2\pi(1-\cos\theta)$

Thus net outflowing momentum / unit time, $f' = \frac{P_e}{c} = \frac{P}{2c}(1-\cos\theta)$ ④

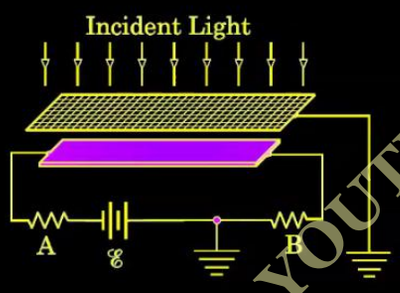
Thus net force on the lens: $F = f' - f = \frac{P}{4c}[2 - 2\cos\theta - \sin^2\theta]$

Now $\cos\theta = d/\sqrt{d^2+r^2}$, $\sin\theta = r/\sqrt{d^2+r^2}$ thus:

$F = \frac{P}{4c}\left[2 - 2d/\sqrt{d^2+r^2} - r^2/(d^2+r^2)\right]$ Ans.

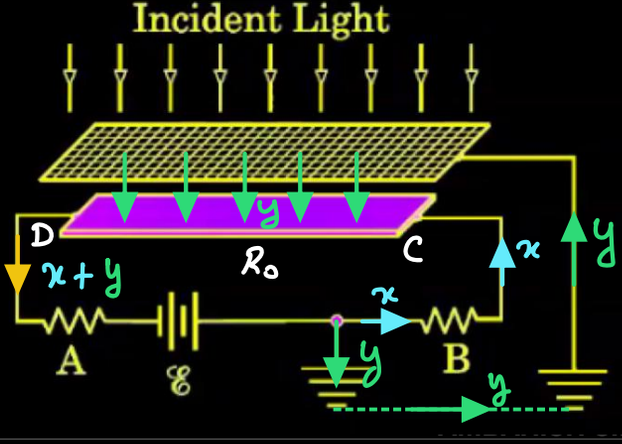
→ Note that momentum flow rate from isotropic source in cone of semi-vertex angle θ: $f = \frac{P}{4c}\sin^2\theta$.

EXAMPLE: PATHFINDER: BUILD 4:



A plate of area S connected to a battery of electromotive force $\mathscr{E}$ through two identical resistors A and B each of resistance R is uniformly illuminated by a monochromatic light of wavelength λ. The photo-electrons emitted from the plate are collected by an earthed wire grid as shown in the figure. Intensity of the light that reaches the plate after passing through the grid is I_0. Resistance of the plate to current flow along its length is R_0. Quantum efficiency of the photoelectric emission is η. Find the strength of currents through each of the resistors A and B. Denote modulus of charge on an electron by e and the Planck's constant by h.

Solution: Here due to photo emission from the plate, the current in the plate CD is not constant, rather it increases as one moves from C to D (more and more electrons are emitted as one moves from C to D which are replenished from earth).



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The equivalent circuit is shown in the figure above. Since photocurrent is uniformly distributed, for purpose of calculating potential drop between C and D we can assume an average current of $x + y/2$. Now applying KVL along loop ABCDA:

$$(x+y)R - \mathcal{E} + xR + (x+y/2)R_0 = 0 \quad \text{①}$$

Also using Photoelectric efficiency and intensity, photocurrent:

$$y = \frac{I_0 S}{(hc/\lambda)} \times \eta e = \frac{\lambda I_0 S \eta e}{hc} \quad \text{②}$$

Using ② in ① and solving for x, $x = \mathcal{E}/(R_0+2R) - y/2$

Thus current through $B = x = \mathcal{E}/(R_0+2R) - (\eta e \lambda I_0 S)/2hc$

Current through $A: x+y = \frac{\mathcal{E}}{R_0+2R} + (e\eta\lambda I_0 S/2hc)$ Ans.

EXAMPLE: PATHFINDER: MCQ:10-12:

A narrow beam of monochromatic light of wavelength λ emitted from a source of power P is propagating in the positive x-direction. After being reflected from a perfectly reflecting plane mirror of area vector $\vec{A} = A(-\hat{i} + \hat{j})$, the beam falls on a metal plate of surface area A placed parallel to the x-z plane. Work function of the metal is ϕ and photoelectric efficiency is η. Photoelectrons emitted from the metal plate are immediately removed from the vicinity of the plate.

10. Reflection from the mirror changes momentum of photons and exerts force on the mirror. Which of the following statements is/are correct?

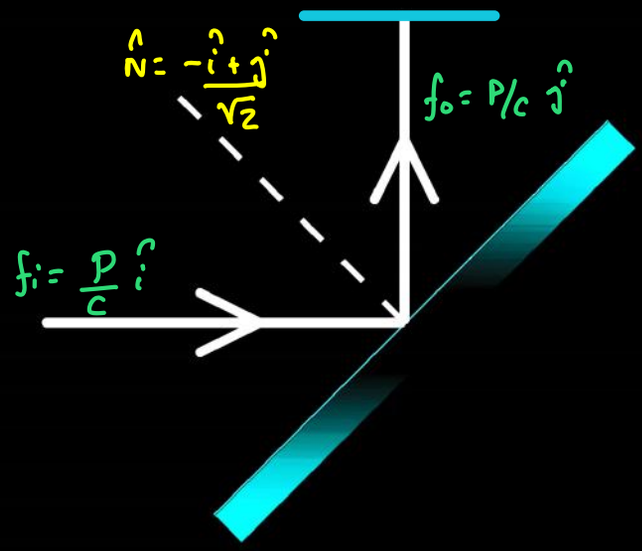
(a) Change in momentum of each photon is $\frac{h}{\lambda}\left(-\hat{i} + \hat{j}\right)$.

(b) Change in momentum of each photon is $\frac{h}{\lambda}\left(\hat{i} - \hat{j}\right)$.

(c) Force exerted by light beam on the mirror is $\frac{P}{c}\left(\hat{i} - \hat{j}\right)$.

(d) Force exerted by light beam on the mirror is $\frac{P}{c}\left(-\hat{i} + \hat{j}\right)$.

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Solution: The ray diagram for the situation is shown in the adjacent figure. If P is the power of the source, Momentum flow rate, $f = P/c$. Thus incoming momentum flow rate, $f_i = P/c\ \hat{i}$ and outflowing momentum rate $f_o = P/c\ \hat{j}$. Now force on the light beam by the mirror, $\vec{f}_e = \vec{f}_o - \vec{f}_i$ and its reaction acts on the mirror thus $\vec{f}_m = \vec{f}_i - \vec{f}_o$ i.e $\frac{P}{c}(\hat{i}-\hat{j})$ ∴ (c) is correct.

Now change in momentum of each photon:

$\Delta\vec{p} = \frac{h}{\lambda}\hat{j} - \frac{h}{\lambda}\hat{i} = \frac{h}{\lambda}(-\hat{i}+\hat{j})$ thus (a) is correct.

Thus correct options are (a) and (c).

11. After a time interval Δt from the instant light falls on the metal plate, electric field at a point very close to the metal plate and far away from its edges is explored. Which of the following statements is/are true

(a) Modulus of electric field is $\frac{\eta P \lambda e \Delta t}{4\varepsilon_0 hcA}$.

(b) Modulus of electric field is $\frac{\eta P \lambda e \Delta t}{2\varepsilon_0 hcA}$.

(c) Modulus of electric field is $\frac{\eta P \lambda e \Delta t}{\varepsilon_0 hcA}$.

(d) Electric field increases linearly with time in starting and reaches saturation after a short while.

Solution: Photoelectric current, $i = \frac{P}{hc/\lambda} \times \eta e = \frac{\eta P \lambda e}{hc}$

Now electric field: $E = \sigma/2\varepsilon_0 = i\Delta t/2\varepsilon_0 A = \frac{\eta P \lambda e \Delta t}{2hc\varepsilon_0 A}$

option (b) shows.

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12. If power of the source, wavelength of light, surface area and photoelectric efficiency of the metal of the plate are adjusted to 60 W, 6000 Å, 0.04 m^2 and 10^{-4} respectively. Which of the following statements are correct? Use $hc = 12420$ eV·Å.

(a) Range of kinetic energy of photoelectrons does not depend on photoelectric efficiency of the metal used.

(b) If work function of the metal of the ball is 1.9 eV, the maximum kinetic energy of the photoelectrons is 0.17 eV.

(c) If material of metal plate has work function 2.1 eV, no photoelectrons are observed.

(d) Number of photoelectrons emitted per second is 1.82×10^{16}.

Solution: (a) Range of KE is from 0 to K_{max} which is only dependent on frequency and work function $\therefore$ (a) is obviously correct

(b) $K_{max} = h\nu - \phi = \frac{hc}{\lambda} - \phi = \frac{1242\ eV\,nm}{600\ nm} - 1.9\ eV \approx 0.17\ eV$

Thus (b) is correct

(c) hc/λ is $(1242/600)\,eV = 2.067\ eV$, this is less than ϕ (i.e 2.1 eV) so no emission will occur $\therefore$ (c) is also correct

(d) Now $n_e = \left(\frac{P}{hc/\lambda}\right) \times \eta = \frac{P\lambda}{hc} \times \eta = \frac{60 \times 600\ nm \times 10^{-4}\ W}{1242\ eV\,nm}$

$= \frac{60 \times 600}{1242 \times 1.6 \times 10^{-19}}\ /s = 1.81 \times 10^{16}/s$

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